

Kiran Baby

[Website](#) | [GitHub](#) | [LinkedIn](#) | [Dev.to](#)
kiranbaby256@gmail.com | +447767977157

Experience

Moonbility, *London, UK*

Founding Engineer (Aug 2024 – Present)

Live Bus Network Monitoring & Intervention Platform

Core Architect & Developer

In production: Greater Manchester

- Architected and built the core platform end-to-end (backend, data pipeline, operator-facing UI) for monitoring **1,600+ live vehicles** across a major UK metropolitan bus network; used by control room operators to resolve **300+ operational issues** in live deployment ([EIT Urban Mobility coverage](#))
- Designed multi-strategy matching engine that correlates real-time SIRI-VM vehicle feeds against scheduled timetable data using multiple identifier fallbacks, time matching, and calendar-aware filtering
- Built real-time progress and delay system with GPS-based stop-arrival detection, bearing validation, circular-route handling, and GPS-gap handling to avoid false delay readings across skipped stops
- Developed disruption impact analysis tool allowing control room operators to draw areas on a map and instantly see affected services, imminent vehicles, and historical delay patterns
- Implemented route intervention workflow (diversion simulation with real-time traffic speeds, short-turn analysis, shareable proposals with approval chain), demonstrated on-site at the regional transport control room
- Engineered change-detection pipeline polling GPS feeds on a 2s interval, recording thousands of vehicle positions per cycle to PostGIS with Redis caching, WebSocket broadcast, and in-memory state synchronisation
- Built incremental TransXChange ingestion for 2,000+ daily timetable files with multi-layer change detection (ETag, revision number, SHA-1 hash), enabling 6-hourly cron refreshes without reprocessing the full dataset
- Collaborated with control room operators and company leadership through on-site workshops, translating operational needs into shipped features

Python, FastAPI, PostgreSQL/PostGIS, Redis, WebSockets, Next.js, Mapbox, SQLAlchemy

Transport Monitoring CV System, MVP (ITS UK Award Winner)

Full-Stack & CV Engineer

- Engineered passenger counting and bay-area monitoring system with parallel video stream processing via Python multiprocessing, a YOLO-based detection pipeline combining multiple models, and Redis for cross-camera data unification
- Trained and deployed CV models on GCP for inference
- Configured IP cameras and network routing (port forwarding, RTSP stream setup) to establish reliable multi-camera feeds into the pipeline

Python, FastAPI, OpenCV, Next.js, Redis, GCP, Server-Sent Events, YOLOv11

Live Train Tracking & Delay Prediction, MVP

Full-Stack & ML Engineer

- Built ETL pipeline ingesting real-time train movement data from National Rail's Darwin Push Port, trained custom delay-prediction model on historical rail data, and streamed live train positions to frontend over WebSockets

Python, FastAPI, Next.js, Mapbox.js, WebSockets, ML, AWS S3, MongoDB

Westland Technologies, *Glasgow, Scotland*

Software Developer (Oct 2023 – May 2024)

AI Product Recommendation Chatbot

Backend AI Developer

- Developed RAG-based product recommendation chatbot for client with 10k+ monthly users, increasing engagement by 25%; built web scraping pipeline ingesting 400+ product records

Python, LangChain, OpenAI, FAISS, Flask, Scrapy, MongoDB, React

Projects

MiniLondon3D. Real-time 3D London Underground tracker. Python worker polls TfL APIs, estimates train positions, and snaps to track geometry. Decoupled architecture with Redis pub/sub, WebSocket push with REST fallback. Three.js instanced meshes on MapLibre. [Live](#) | [Blog](#)

SearchLightAI. Video moment search engine using hybrid SigLIP2 visual embeddings + Sentence Transformer text embeddings in Qdrant, with PostgreSQL for metadata. PySceneDetect keyframe extraction and faster-whisper GPU transcription. Score normalisation across visual and speech collections. Full stack containerised with Docker Compose and NVIDIA GPU passthrough. [GitHub](#)

TalkMateAI. Real-time AI avatar integrating Whisper STT, SmolVLM2 vision, and Kokoro TTS with word-level lip-sync over WebSocket streaming with chunked audio generation. [GitHub](#)

Awards

- 2025 **Winner**, [ITS UK Better Accessibility through Technology Award](#) – Real-time Onboard Monitoring (Moonbility)
- 2025 **Shortlisted**, [ITS UK Eric Sampson Award for Early Careers Professional of the Year](#)
- 2021 **3rd Prize**, Game Begetter at IIT Bombay Techfest
- 2021 **Finalist**, 10th CSI-InApp International Student Project Awards

Technical Skills

Languages	Python, TypeScript
Databases	PostgreSQL, PostGIS, MongoDB, Redis, Alembic
DevOps/Cloud	Docker, GCP, GitHub Actions
AI/ML	PyTorch, HuggingFace, RAG, Computer Vision
Web	Next.js, React, FastAPI
Tooling	Ruff, Prettier, Husky, pre-commit

Education

MSc in Artificial Intelligence	University of St Andrews, UK (2022–2023) Merit
B.Tech in Computer Science	FISAT, Kerala, India (2017–2021) First-Class Honours